

Exercise 3

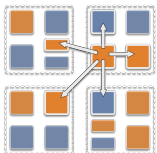
Goal: implement a sorting algorithm based on odd-even swapping

- using Charm4Py **channels**

Odd-even sort: a parallel sorting algorithm that repeatedly compares and swaps adjacent elements in alternating phases:

- Odd phase: Compare and swap elements at indices (1,2), (3,4), (5,6), etc.
- Even phase: Compare and swap elements at indices (0,1), (2,3), (4,5), etc.
- Repeat these phases until the array is sorted

This pattern allows parallel comparisons since non-adjacent pairs are independent.

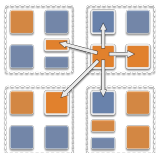


Charm4Py Implementation

- Chare array is constructed such that chares are each assigned a single element.
- **TODO:** every chare establishes **channels** to its left and right neighbors
- **TODO:** chares iteratively swap elements appropriately such that the chare array eventually contains the sorted list.

Every iteration is either odd or even.

- During even phases, even-indexed chares "lead" the exchange, sending their value to the right and waiting on a response.
- During odd phases, odd-indexed chares "lead" the exchange similarly.



Exchange Details

During every phase, the leading chare performs the following:

- send value right
- wait on a response r
- set `self.value = r`

The receiving chare will instead:

- receive a value v from the left
- compare v with `self.value`
 - If $v \leq \text{self.value}$ then the pairwise ordering is acceptable. Respond with v .
 - Otherwise, the pairwise ordering must be corrected. Respond with `self.value` and update `self.value = v`.

